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Engineering the Baseline Resistance of Al-Doped ZnO Thin Films for Chemiresistive Gas Sensor Platforms

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Abstract

Al-doped zinc oxide (AZO) thin films were deposited by RF magnetron sputtering using a simple doping approach: aluminum tape was placed directly on the ZnO target. The work focused on understanding how Al incorporation, annealing temperature, and annealing atmosphere affect the structural and electrical behavior of the films, particularly their baseline resistance for gas sensing applications. EDX measurements confirmed that increasing the Al-covered area on the target progressively increased the Al concentration in the deposited layers. XRD analysis showed that higher Al contents and stronger thermal treatments reduced crystallinity and promoted the formation of smaller crystallites. The electrical response changed markedly with both Al incorporation and thermal treatment. Pure ZnO films initially exhibited very high resistance, while annealing reduced it by several orders of magnitude. Further increasing the Al concentration improved conductivity due to the donor effect of Al in the ZnO lattice. Overall, the results show that RF sputtering and post-treatment strategies can provide broad control over the electrical resistance of ZnO-based thin films. This tunability makes AZO coatings attractive for adapting chemoresistive gas sensors to different sensing environments and operating regimes, and deposition and thermal treatment parameters determine the resistance range.

Keywords: sheet resistance; biosensors; annealing; X-ray diffraction

1. Introduction

Gas sensors play an increasingly important role in modern society, with applications spanning environmental monitoring, industrial safety, and medical diagnostics [1]. They are widely used for detecting hazardous gases such as methane in mining environments, monitoring vehicle emissions, and assessing indoor air quality [1,2]. More recently, gas-sensing technologies have also attracted attention in the medical field, where sensors are used to detect biomarkers in exhaled breath for disease diagnosis [3]. In addition, the development of wearable gas sensors has opened new possibilities in areas such as animal health monitoring and precision livestock farming [1,4–6]. Table 1 summarizes representative chemoresistive gas-sensing systems and their preferred resistance ranges, depending on the dominant surface reaction mechanism and sensing conditions.

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Table 1. Characteristics of common chemoresistive gas sensors.

Target Gas/Sensing Mechanism	Typical Surface Reaction or Sensing Feature	Common Material Systems	Preferred Resistance Range
Hydrogen (H_2) sensing [7]	$H_2 + O^-$ ads $\rightarrow H_2O + e^-$	Pd-doped ZnO, SnO_2	Low–moderate 10^2–10^5 Ω
Carbon monoxide (CO) sensing [8]	$CO + O^-$ ads $\rightarrow CO_2 + e^-$	SnO_2, ZnO	Moderate 10^3–10^6 Ω
Methane and combustible gas sensing [9]	Catalytic oxidation reactions at elevated temperatures	WO_3, Pt/Pd-modified oxides	Low–moderate 10^2–10^6 Ω
NO_2 sensing [10]	$NO_2 + e^- \rightarrow NO_2^-$	WO_3, ZnO nanowires	High resistance 10^6–10^{10} Ω
VOC sensing (acetone, ethanol, toluene, formaldehyde) [11]	Weak or moderate adsorption-induced charge transfer	ZnO, In_2O_3	Moderate–high 10^5–10^9 Ω
Room-temperature or UV-assisted sensing [12]	Photoactivated surface reactions with low carrier density	AZO, graphene/oxide hybrids	High resistance 10^6–10^{10} Ω

Although the optimal resistance window strongly depends on the target gas and sensing conditions, these differences primarily reflect the extent of charge transfer during gas–surface interaction. Reducing gases such as H_2 or CO typically promotes strong electron release into n-type metal oxides, allowing stable operation at relatively low resistance values [7,8]. In contrast, oxidizing gases such as NO_2 or low-concentration VOC detection often benefit from higher baseline resistance, where small variations in surface depletion can produce larger relative electrical changes [9–11]. Thus, the electrical resistance of ZnO-based transducers is not only determined by the intrinsic material properties but also represents a key design parameter that influences sensitivity, signal stability, and practical sensor operation [8]. Recent studies on low-resistance gas sensors have also highlighted the importance of engineering the baseline resistance of conductive oxide films to achieve stable sensor operation and facilitate signal readout [13].

A variety of gas sensing technologies have been developed, including electrochemical sensors, catalytic bead sensors, photoionization detectors, infrared absorption sensors, and semiconductor-based sensors [2,14,15]. Semiconductor metal oxide (SMO) sensors have received significant attention due to their simple design, low cost, and high sensitivity [2,15,16]. These sensors operate by detecting changes in electrical properties resulting from interactions between gas molecules and the sensor surface. Depending on the dominant charge carriers, SMO sensors are classified as either n-type or p-type semiconductors. N-type materials, such as zinc oxide (ZnO) and tin oxide (SnO_2), primarily conduct through electrons, while p-type materials conduct via holes [17,18]. The adsorption and reaction of gas species on the surface modify the concentration of charge carriers, leading to measurable changes in electrical resistance [19].

Thus, zinc oxide is considered one of the most promising materials for gas-sensing applications due to its wide bandgap, high electron mobility, chemical stability, and low toxicity [6,20]. In addition, ZnO is relatively inexpensive and can be synthesized using a wide range of deposition techniques, making it highly suitable for large-scale applications [6,17,21–23].

ZnO-based sensors have demonstrated sensitivity to various gases, including hydrogen, ammonia, hydrogen sulfide, nitrogen dioxide, and volatile organic compounds [5,22,24–26].

To further enhance the performance of ZnO-based sensors, several strategies have been explored: (i) doping with foreign elements, (ii) controlling the microstructure, and (iii) optimizing thin-film properties [27]. Among these approaches, aluminum doping has

attracted considerable interest, as it can improve electrical conductivity and modify surface properties, thereby enhancing sensing performance [28–31].

Al-doped ZnO (AZO) thin films have been researched for sensors in various gas sensing applications since they represent a promising material system for health and environmental applications [4,25,29,30]. However, achieving a resistance level that matches the requirements of a specific sensing application remains a practical challenge. In this work, AZO thin films were deposited by RF magnetron sputtering using different aluminum contents, deposition temperatures, and post-deposition annealing conditions to examine their influence on the electrical resistance. Based on these results, the study identifies processing routes that enable resistance tuning over a broad range and discusses their relevance for the design of chemiresistive gas sensors requiring different baseline resistance windows.

2. Experimental Section

2.1. Substrates and Device Preparation

Three types of substrates were used for thin-film deposition: glass slides, silica substrates, and microfabricated gas-sensing devices. The sensing devices served as functional samples for electrical characterization, while glass and silica substrates were used as reference samples for structural and compositional analyses. Gas sensing characterization was beyond the scope of the present work, which was designed to establish the relationship between processing parameters, microstructure, and baseline electrical resistance.

The sensing devices had approximate dimensions of 10 mm × 6 mm and featured interdigitated electrodes on their surface. During deposition, the thin film was applied to the active region between the electrodes, forming a conductive pathway across the device. The device and the mask holder used are shown in Figure 1a,b. This configuration enabled direct measurement of changes in the film's electrical properties through the electrode contacts. Two deposition approaches were employed. In the initial experiments, the film covered a relatively large area of the device surface. In subsequent experiments, the deposition area was confined to the active region using a polyimide mask, ensuring that the film was restricted to the sensing zone. The device's electrode spacing configuration was 5 µm.

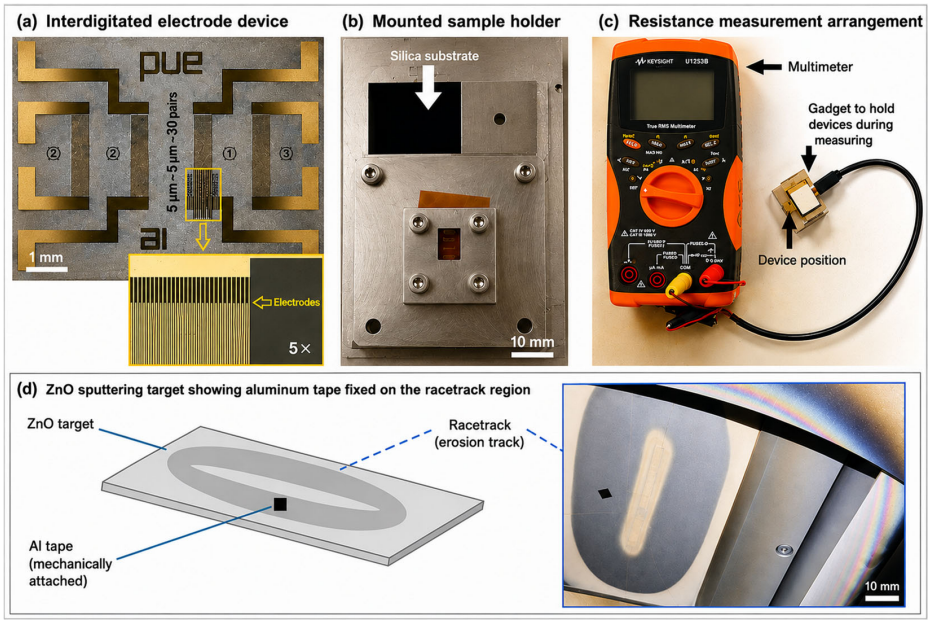


Figure 1. (**a**) Photographs of a gas sensing device, (**b**) substrate holder and mask prepared for the deposition of the device, (**c**) gadget and multimeter used for sheet resistance measurement, and (**d**) photograph of the ZnO sputtering target showing the aluminum tape mechanically fixed on the racetrack region prior to deposition.

Prior to deposition, the glass and silica substrates were cleaned with isopropanol to remove surface contaminants and then dried. The sensing devices were cleaned using compressed gas to eliminate dust particles without damaging the electrode structures.

2.2. RF Magnetron Sputtering

ZnO and Al-doped ZnO (AZO) thin films were deposited by RF magnetron sputtering using a Balzers BAS 450 sputtering system equipped with rectangular ZnO targets. Depositions were carried out under an Ar atmosphere with a gas flow rate of 75 sccm and an RF power of 200 W. The chamber base pressure was in the range of 10^{-6}–10^{-7} mbar, while the operating pressure was approximately 0.03 mTorr and the deposition time was typically 30 min. In this study, the deposition temperature refers to the nominal substrate-holder temperature set during sputtering. It was monitored using the thermocouple integrated into the substrate heater assembly, with the sensor located behind and in close thermal contact with the substrate holder. No direct measurement was performed on the substrate surface. Different Al tape areas were used to systematically vary the Al concentration in the deposited films. Metallic Al tape pieces, ranging from 72 to 912 mm^2, were mechanically fixed onto the racetrack region of the ZnO target, as shown in Figure 1d. This approach locally modified the target composition and erosion characteristics, enabling systematic control of Al incorporation while maintaining stable plasma conditions during sputtering. As shown in Table 2, the experiments were divided into three groups focused on (i) evaluating the effect of deposition temperature, (ii) varying Al incorporation, and (iii) studying the influence of the annealing temperature on AZO. The remaining sputtering conditions were kept constant throughout the study.

Table 2. Deposition and post-deposition treatment experiments.

Experimental Series	Variable	Values
ZnO films	Deposition temperature	RT, 150, 175, 200, 225 °C
AZO films	Al concentration	0.5, 0.7, 0.8, 1.0, 1.3 at.%
Annealed AZO films	Annealing temperature	Not annealed, 150, 300, 450 °C (1 h, vacuum)

2.3. Post-Deposition Annealing and Atomic Force Microscopy

AZO with 0.5 Al Me at. % samples were heated for one hour at the desired temperature under vacuum and then allowed to cool naturally to room temperature. The maximum annealing temperatures investigated were 150, 300, and 450 °C; the average heating rate was 3 °C/min. Annealing was performed only on the 0.5 at.% AZO films because this study aimed to achieve controlled tuning of the baseline resistance rather than its maximum reduction. The intermediate Al concentration provided a suitable starting point to investigate the specific effect of thermal treatment while minimizing the influence of simultaneous changes in dopant content. Since the original experiments did not include independent recording of the cooling ramps, these rates are not reported here. Surface morphology characterization was performed using a Bruker Dimension Icon atomic force microscope. AFM images were acquired in tapping mode under ambient conditions and processed using NanoScope Analysis software (version 1.85) and JPK SPM Data Processing software (version 8.0.13). The measurements were used to evaluate the evolution of surface roughness, grain size, homogeneity, and morphological defects in the ZnO and AZO thin films following thermal treatment.

2.4. Electrical Resistance Measurement

Electrical resistance measurements were performed on the sensing devices after deposition and thermal treatment. The reported resistance values are the average of measurements taken between the four electrode contacts on each device.

Measurements were carried out using a (Keysight Technologies, Santa Rosa, CA, USA; model U1241B) capable of measuring resistances up to 100 MΩ. A dedicated connector shown in Figure 1 c was used to interface the electrodes with the measurement system. All measurements were taken after the samples had cooled to room temperature to ensure consistency.

2.5. Grazing Incidence X-Ray Diffraction

The crystalline structure of the films was analyzed using grazing incidence X-ray diffraction (GIXRD). This technique is particularly suitable for thin films, as the low incident angle limits the penetration depth of X-rays and minimizes the contribution from the substrate.

Measurements were performed using a Siemens D5005 diffractometer (Siemens AG, Munich, Germany) equipped using DIFFRAC.EVA software (Version 4.2, Bruker AXS GmbH, Karlsruhe, Germany). Diffraction patterns were collected over a 2θ range of 28° to 64°, with a step size of 0.04° and a counting time of 2 s per step. A fixed incident angle of 0.5° was used throughout the analysis.

2.6. Energy-Dispersive X-Ray Spectroscopy

The elemental composition of the films, particularly the aluminum metal atomic percent content (denoted as Me at. %), was determined using energy-dispersive X-ray spectroscopy (EDX) to quantify the atomic percentage of aluminum and zinc in the samples. The morphology was examined using a FEI XL30 scanning electron microscope (FEI Company, Hillsboro, OR, USA) equipped with an energy-dispersive X-ray spectroscopy (EDS) detector (Oxford Instruments plc, Abingdon, UK).

3. Results

3.1. Microstructure and Elemental Distribution of Pure ZnO

The cross-sectional morphology and elemental distribution of the ZnO films were examined using SEM and EDX mapping, as shown in Figure 2. The SEM image (Figure 2a) clearly distinguishes the ZnO layer from the underlying Si substrate, confirming the formation of a well-defined film.

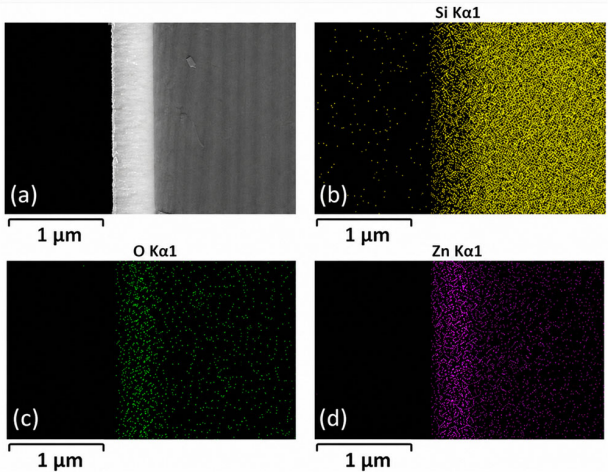


Figure 2. (**a**) Cross-sectional SEM image of the ZnO thin film deposited on a Si wafer. (**b**) EDX elemental map of Si (Si–Kα), highlighting the substrate region. (**c**) EDX elemental map of oxygen (O–Kα). (**d**) EDX elemental map of zinc (Zn–Kα). The elemental maps confirm the spatial distribution of the film constituents across the cross-section.

Elemental mapping further provides insight into the spatial distribution of the constituent elements. The Si signal (Figure 2b) is confined to the substrate region, while the Zn and O maps (Figure 2c,d) show a uniform distribution across the entire thickness of the film. This homogeneous distribution indicates that the sputtering process produces compositionally consistent ZnO layers without noticeable segregation or compositional gradients.

3.2. Aluminum Incorporation and Compositional Analysis

Aluminum incorporation into the ZnO thin films was controlled by varying the area of Al tape positioned within the sputtering target racetrack region. EDX analysis confirmed that increasing the Al-covered area resulted in progressively higher Al concentrations in the deposited films. For the lowest Al coverage, the measured concentration remained below approximately 0.5 Me at.%, while the largest Al-covered areas produced concentrations above 1 Me at.%. This behavior indicates that the Al tape approach provides a simple and effective method for tuning dopant incorporation during sputtering. Placing the Al pieces directly within the racetrack region was found to promote the efficient transfer of Al into the growing ZnO layer.

3.3. Structural Properties

The crystallographic properties of the films were analyzed using XRD, as shown in Figure 3.

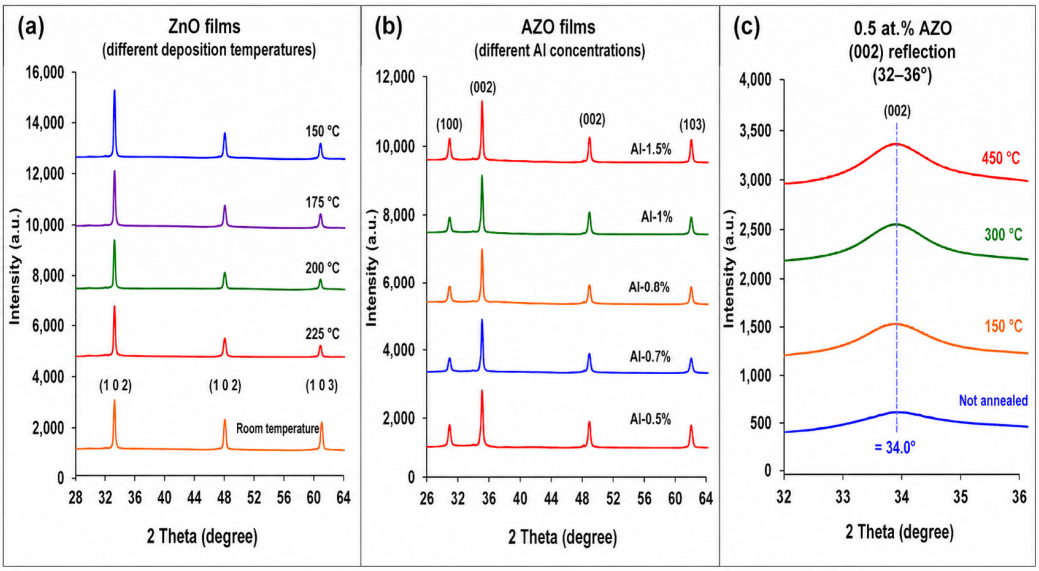


Figure 3. (**a**) X-ray diffraction (XRD) patterns of pure ZnO thin films deposited at different deposition temperatures, (**b**) XRD patterns of Al-doped ZnO (AZO) thin films with varying Al metal concentrations, (**c**) 0.5 at. % AZO annealed at various temperatures.

For pure ZnO films, the dominant diffraction peak corresponds to the (002) plane of the hexagonal wurtzite structure. However, the intensity of this peak gradually decreases with increasing deposition temperature. This behavior indicates a slight reduction in crystallinity at deposition temperatures above 150 °C. Such changes may be attributed to the release of internal stress in RF-deposited films [32,33].

Figure 3b shows the XRD patterns of Al-doped ZnO films with varying Al concentrations. The pure ZnO film exhibits a significantly higher (002) peak intensity at approximately 34.6°, indicating larger crystallite size and better crystallinity. The Al-doped films exhibit lower peak intensities than pure ZnO films deposited at 150 °C, suggesting the formation of more polycrystalline structures.

One possible explanation is that Al incorporation modifies the early stages of film growth and may promote the formation of a larger number of nuclei, leading to smaller crystallites and reduced preferential orientation. Similar trends have been reported in previous studies of AZO thin films. As a result, a larger number of smaller crystallites are formed instead of fewer larger ones. The main diffraction peaks for the doped films appear at approximately 33.56°, 33.72°, 33.76°, and 33.72° for Al concentrations of 0.5%, 0.8%, 1%, and 1.3%, respectively. These peaks correspond to a shifted (002) ZnO reflection from the standard position at 34.364°, rather than the (100) ZnO or (104) Al_2O_3 peaks. In Figure 3c, only the 2θ range from 32° to 36° is presented, as no significant peaks were observed outside this region for Al 0.5 at. % AZO films deposited at room temperature. The (002) peak loses intensity and becomes more amorphous after annealing. Interestingly, AFM analysis (Figure 4) shows a concurrent increase in grain size with annealing temperature. This indicates that thermal treatment promotes surface grain coalescence while reducing the degree of preferential (002) crystallographic orientation detected by XRD.

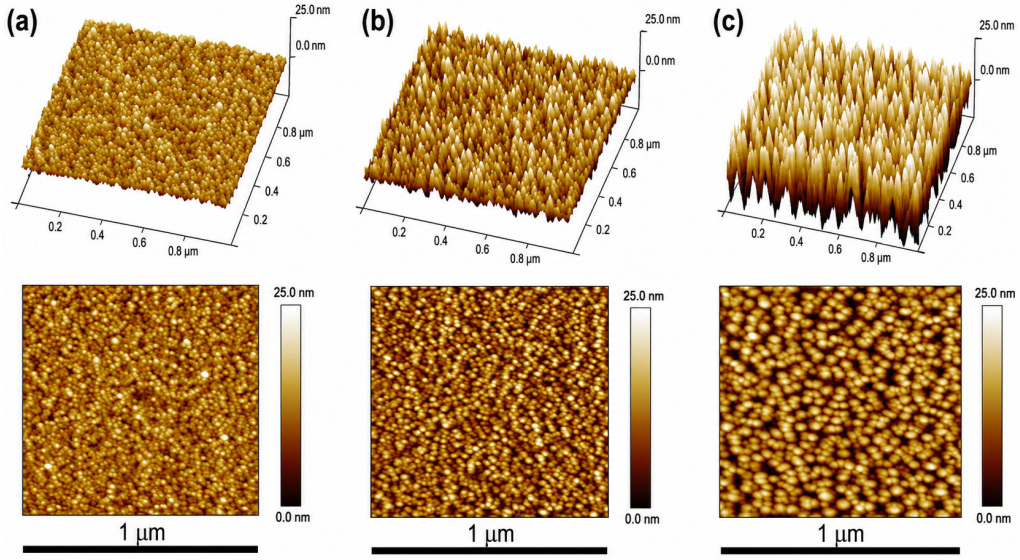


Figure 4. Atomic force microscopy (AFM) images of AZO thin films annealed for 60 min at: (**a**) unannealed, (**b**) 150 °C, (**c**) 450 °C. The top row shows the three-dimensional surface morphology, while the bottom row presents the corresponding two-dimensional height maps, illustrating the evolution of surface texture with annealing time.

3.4. Surface Morphology Evolution

The effect of annealing temperature on the surface morphology of the 0.5 Al at.% AZO films presented were investigated using AFM, as presented in Figure 4. The results demonstrate that annealing significantly influences surface topography.

The AFM images demonstrate a clear change in the surface structure of AZO thin films as the annealing temperature increases. The unannealed sample in Figure 4a shows a comparatively smooth and uniform surface with small, evenly distributed grains. After annealing at 150 °C, shown in Figure 4b, the grains appear more defined and closely packed, suggesting the beginning of grain growth and improved surface organization. In contrast, the film annealed at 450 °C in Figure 4c exhibits larger grain clusters and a more pronounced surface texture, leading to higher roughness and greater height variation across the film surface. The corresponding 2D height maps support these observations by revealing a gradual increase in grain size and surface irregularity with rising annealing temperature. These changes indicate enhanced atomic mobility and grain coalescence during thermal treatment, which may improve the structural and functional properties of the AZO thin films.

3.5. Electrical Properties of ZnO/AZO Films

The electrical resistance of ZnO films as a function of deposition temperature is shown in Figure 5a. A substantial decrease in resistance is observed with increasing deposition temperature, indicating improved electrical conductivity.

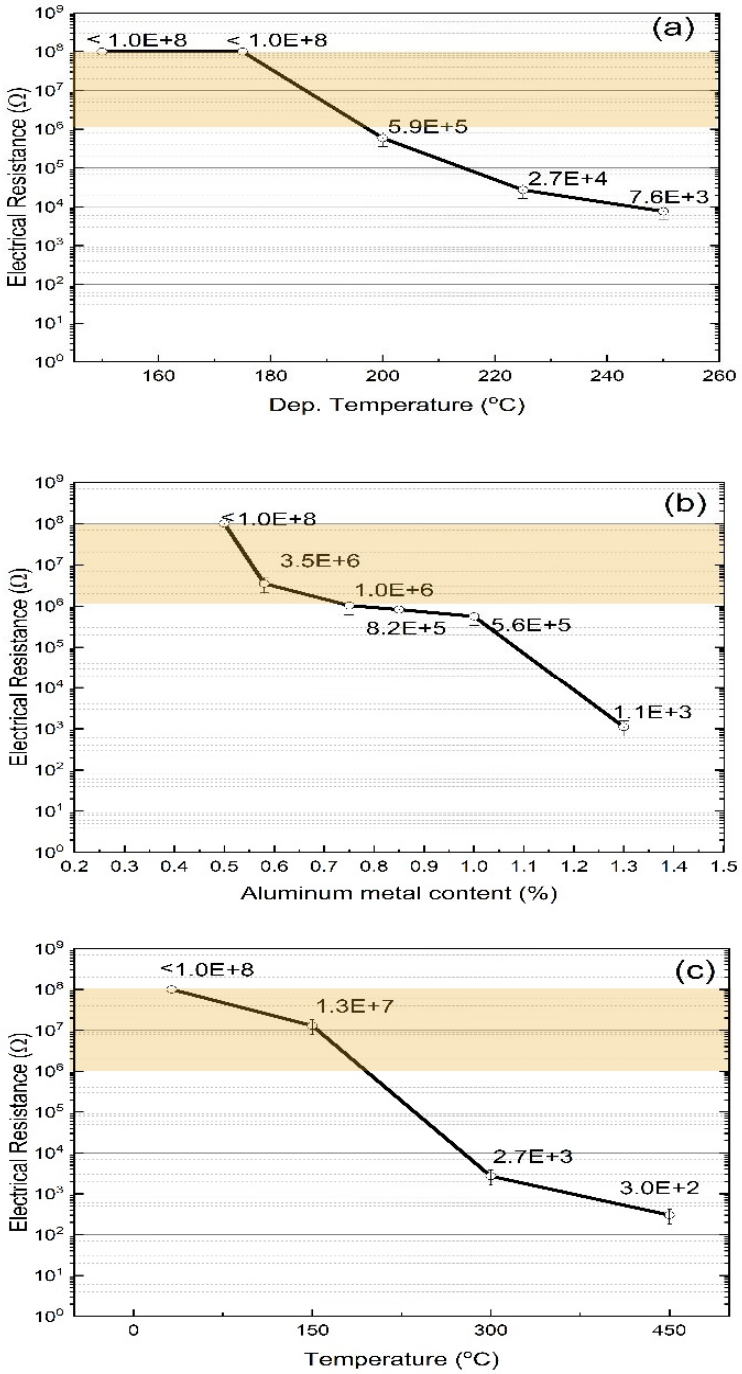


Figure 5. Electrical resistance tuning in ZnO and Al-doped ZnO (AZO) thin-film transducers deposited by RF magnetron sputtering. (**a**) Effect of deposition temperature on pure ZnO films. (**b**) Effect of Al incorporation on AZO films. (**c**) Influence of annealing on 0.5 Al Me at.% AZO films.

The resistance of the ZnO samples deposited at 150 °C and 175 °C exceeds 100 MΩ, which is beyond the instrument's measurement range, reflecting poor charge transport due to low crystallinity and a high density of defects or grain boundaries. Above 175 °C, the resistance decreases below 1 MΩ. A more pronounced reduction occurs at deposition temperatures of 225–250 °C, where the resistance drops to the kΩ range.

This trend suggests that annealing enhances charge transport, likely through improved crystallinity, increased carrier mobility, and better grain-to-grain connectivity [34]. Defect activation and reduction in grain boundary scattering may also contribute to the observed behavior. Figure 5b shows the variation in electrical resistance with Al concentration. In general, the resistance decreases with increasing Al content, demonstrating the effectiveness of Al doping in enhancing the conductivity of ZnO films. At lower Al concentrations (0.5–0.75%), the resistance remains relatively high, indicating limited carrier concentration. A sharp decrease in resistance is observed at 0.8% Al, suggesting a significant increase in free-carrier density due to the substitutional incorporation of Al^{3+} into Zn^{2+} lattice sites. Although a slight increase in resistance is observed at 1.0% Al, the overall trend continues downward, with the lowest resistance achieved at 1.3% Al. This behavior indicates that increasing Al content enhances donor concentration, thereby improving electrical conductivity. However, local variations may arise due to competing effects such as defect formation or dopant clustering at higher concentrations. The influence of the annealing atmosphere on electrical properties is shown in Figure 5c for 0.5 at. % Al-doped ZnO films. Interestingly, the strongest decrease in resistance occurred between 150 and 300 °C, suggesting that this temperature interval may correspond to the activation of the dominant transport mechanisms in the films.

4. Discussion

The present results show that the electrical behavior of ZnO and AZO thin films can be widely adjusted through relatively simple modifications of the deposition and post-treatment conditions. In particular, both Al incorporation and thermal annealing strongly affected the baseline resistance of the films, a key parameter in chemoresistive gas sensors because it influences signal stability, noise levels, and sensitivity to different gases [8,11]. Similar considerations have also been reported for room-temperature gas sensors, where the initial resistance strongly affects the measurable response and practical operation of the sensing platform [35].

Among the evaluated parameters, deposition temperature showed the greatest effect on resistance. The RT-deposited ZnO films exhibited resistance values beyond the instrument's measurable range, suggesting limited carrier transport in the initially sputtered layers. After annealing, the resistance progressively decreased by several orders of magnitude. Similar behavior has been observed in thermally treated ZnO systems, where annealing promotes grain coalescence, improves grain connectivity, and activates donor-like defects associated with oxygen vacancies, even as the degree of preferential crystallographic orientation decreases [32,33]. In the present case, the reduction in resistance after annealing is therefore likely associated with improved carrier mobility and enhanced electrical pathways between neighboring grains.

The morphological observations support this interpretation. AFM analysis revealed noticeable changes in surface texture after thermal treatment. Annealing at 150 °C generated a more homogeneous surface, whereas annealing at 450 °C promoted grain coarsening and increased roughness. This suggests that moderate thermal exposure favors surface diffusion and structural rearrangement, whereas excessive annealing may induce localized grain growth or surface instability. Such behavior has also been observed in oxide thin films where prolonged heating alters the balance between densification and grain coarsening [32]. Since electron transport in polycrystalline ZnO strongly depends on grain boundary continuity, these microstructural changes are expected to directly influence conductivity. Similar relationships between microstructure, charge transport pathways, and impedance response have been observed in hierarchical porous gas-sensing materials, where grain connectivity and defect structure play an important role in determining electrical behavior [36]. The

simultaneous increase in grain size observed by AFM and the decrease in (002) peak intensity observed by XRD suggest that annealing primarily promotes grain coalescence and surface restructuring rather than improving long-range crystallographic order. Although the XRD results show a reduction in (002) peak intensity after annealing, the electrical conductivity increased significantly. This suggests that conductivity is not controlled solely by crystallinity. Annealing likely promotes grain coalescence and improves electrical pathways between grains, while also increasing the activation of donor-like defects such as oxygen vacancies. The AFM results, which show larger grains after annealing, support this interpretation.

Al incorporation also had a marked effect on the electrical properties of the films. In general, increasing the Al concentration lowered the resistance, confirming the donor role of Al in ZnO-based systems [30]. When Al^{3+} substitutes Zn^{2+} lattice positions, additional free electrons are introduced into the material, increasing carrier concentration. However, the resistance evolution was not perfectly linear with dopant concentration. Small deviations observed at intermediate Al contents indicate that conductivity is influenced not only by carrier density but also by structural disorder and defect formation. At higher dopant concentrations, excessive Al incorporation can distort the ZnO lattice or generate defect clusters that partially reduce carrier mobility despite the higher electron concentration [33].

The novelty of this work extends beyond the use of Al tape as a simple dopant source. Unlike most AZO studies that focus on maximizing conductivity or sensing response, this study demonstrates how baseline resistance can be deliberately tuned over several orders of magnitude through Al incorporation and thermal treatment. An additional aspect emerging from the present results is the possibility of defining a simple parameter to describe how well-suited a given ZnO or AZO film is for a specific gas-sensing regime based on its baseline resistance. Since chemoresistive sensors frequently operate across several orders of magnitude in resistance, direct comparison between materials can become difficult when operating conditions are considered. Thus, an engineered baseline resistance index (*EBRI*) is here proposed as a preliminary resistance-matching descriptor to aid in selecting candidate deposition parameters for future gas-sensor validation. Its predictive value must still be confirmed through controlled gas-sensing measurements. The proposed *EBRI* may be expressed as:

$$EBRI = |\log_{10}(R_b) - \log_{10}(R_{target})| \quad (1)$$

where R_b corresponds to the measured baseline resistance of the film and R_{target} represents the preferred resistance associated with a specific sensing mechanism or gas detection regime. Because the comparison is performed on a logarithmic scale, the parameter is better suited to chemoresistive systems, where resistance commonly varies over several decades. In this formulation, lower *EBRI* values indicate that the film's electrical behavior is closer to the desired sensing window. An *EBRI* greater than 1 would indicate poor matching.

The experimental results obtained in this work also suggest that baseline resistance cannot be considered independently from the processing conditions used to obtain the films. Both Al incorporation and annealing strongly influenced the final electrical response. For this reason, a process-aware version of the parameter, referred to here as *EBRIp*, may also be considered:

$$EBRI_p = |\log_{10}\left(\frac{R_b}{R_{target}}\right)| + \lambda\left(\frac{C_{Al}}{C_{Al,max}}\right) + \mu\left(\frac{T_a}{T_{a,max}}\right) \quad (2)$$

In this expression, C_{Al} is the aluminum concentration in the film, $C_{Al,max}$ is the maximum Al concentration considered in the experimental series, T_a is the annealing temperature, and $T_{a,max}$ is the maximum annealing temperature used during processing. The

parameters λ and μ would serve as weighting factors for dopant incorporation and thermal treatment intensity, respectively.

This extended formulation introduces the idea that an optimized sensing layer should not only exhibit resistance within the desired operating window but ideally achieve it with moderate dopant incorporation and limited thermal processing. Although the present work does not attempt to establish universal weighting values for these parameters, the proposed formulation may serve as a first step toward comparing ZnO-based sensing layers fabricated under different processing conditions.

5. Conclusions

ZnO and Al-doped ZnO thin films were successfully deposited by RF magnetron sputtering using a simple Al incorporation strategy based on aluminum tape placed on the ZnO target. This approach enabled progressive control of Al concentration without requiring specialized ceramic targets. Structural characterization showed that both Al incorporation and thermal treatment modified the crystallinity and morphology of the films. Increasing Al content reduced diffraction peak intensity and promoted smaller crystallites, while annealing strongly influenced surface homogeneity and roughness.

Electrical measurements showed that the resistance of the films can be tuned over several orders of magnitude by combining Al doping and annealing. Increasing Al concentration reduced resistance due to the donor behaviour of Al in ZnO, while thermal treatment further modified the electrical response, likely through changes in microstructure and defect states. These results demonstrate that relatively simple deposition and post-treatment strategies can effectively tailor the baseline resistance of ZnO-based thin films. The proposed engineered baseline resistance index (*EBRI*) is intended as a preliminary framework for resistance matching and requires validation through future gas-sensing studies, including assessments of sensitivity, selectivity, stability, and response kinetics.

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Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The author declares no conflicts of interest.

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